

Workshop Sheet - Week 2

Sets, functions and relations in Lean. Below are some of the new tactics we will be using today.

constructor	Split an $\wedge$ goal or prove both directions of $\leftrightarrow$.
ext x	Prove a set equality by comparing membership of x .
change P	Replace the goal by a definitionally equal expression P .
funext x	Prove a function equality by comparing values at x .
calc	Write a chain of equalities, giving a proof for each step.
ring	Prove a polynomial identity by expanding and collecting terms.

It's helpful to know that Lean unfolds definitions automatically, so $x \in S \cap T$ means $x \in S \wedge x \in T$, and $x \in \{y\}$ means $x = y$. The tactic `change` lets you switch between these forms. The notation `fun x => x + 1` defines the function $x \mapsto x + 1$. For $f : X \rightarrow Y$, $S : \text{Set } X$ and $U : \text{Set } Y$, the expression `f ⁻¹ ' U` denotes the preimage $f^{-1}(U)$ and `f '' S` the image $f(S)$.

1. *Set theory.* Let $S, T \subseteq X$, $U \subseteq Y$, $f : X \rightarrow Y$, and $x \in X$.
 - (a) If $x \in S$ and $x \in T$, prove that $x \in S \cap T$.
 - (b) If $x \in S \cap T$, prove that $x \in S$.
 - (c) If $S \subseteq T$ and $x \in S$, prove that $x \in T$.
 - (d) If $f(x) \in U$, prove that $x \in f^{-1}(U)$.
 - (e) If $x \in S$, prove that $f(x) \in f(S)$.
 - (f) Prove that $f(S \cap T) \subseteq f(S) \cap f(T)$.
 - (g) Prove that $f(S \cap f^{-1}(U)) = f(S) \cap U$.
 - (h) Prove that $\mathcal{P}(S \cap T) = \mathcal{P}(S) \cap \mathcal{P}(T)$. Use `Set.powerset S` for the power set of S .
2. *Functions.* Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$.
 - (a) If $g \circ f$ is injective, prove that f is injective. (Hint: `Function.Injective f`)
 - (b) If $g \circ f$ is surjective, prove that g is surjective. (Hint: `Function.Surjective g`)
 - (c) If $r, s : Y \rightarrow X$ satisfy $r \circ f = \text{id}_X$ and $f \circ s = \text{id}_Y$, prove that $r = s$. (Hint: `congrFun h x`)
 - (d) Prove that the map $X \rightarrow \mathcal{P}(X)$ given by $x \mapsto \{x\}$ is injective.
 - (e) Prove and formalize the following theorem.

Theorem (Cantor). *For every set X , there is no surjection $X \rightarrow \mathcal{P}(X)$.*
3. *Relations.* Fix $n \in \mathbb{Z}$ and define $a \sim_n b$ if $n \mid (b - a)$. Write $[a]_n = \{x \in \mathbb{Z} \mid a \sim_n x\}$.
 - (a) Prove that $\sim_n$ is reflexive.
 - (b) Prove that $\sim_n$ is symmetric.
 - (c) Prove that $\sim_n$ is transitive.
 - (d) Prove that $\sim_n$ is an equivalence relation.
 - (e) If $a \sim_n a'$ and $b \sim_n b'$, prove that $a + b \sim_n a' + b'$.
 - (f) Prove that $[a]_n = [b]_n$ if and only if $a \sim_n b$.